

A MIDI-Controlled Water-Droplet Interface for Generating Droplet Impact Sounds

Taisei Kato
Tokyo Metropolitan University
Tokyo, Japan
lack.s.e.66@gmail.com

Tetsuaki Baba
Tokyo Metropolitan University
Tokyo, Japan
baba@tmu.ac.jp



Figure 1: Instrumental Performance System, Droplet Generator, Pitch Resonator (left to right)

Keywords

Experience, Human-Computer Interaction, Music & Sound, Performance/Stage

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1 Introduction

Since ancient times, water sounds have been objects of profound appreciation, as evidenced by the "Suikinkutsu" (water bells) found in traditional Japanese gardens, as well as in modern musical compositions and ASMR content. However, in the context of musical expression, the use of water sounds has predominantly been limited to the playback of sampled recordings. Methods for performing by controlling the physical behavior of water itself in real-time have not been deeply explored. Furthermore, existing decorative water-sound devices lack the bidirectionality required to reproduce specific pitches or rhythms as intended by a performer.

In this paper, we propose an instrumental performance system that uses the sounds generated when water droplets collide with a surface as a sound source. The proposed system achieves musical expression in rhythm, volume, and pitch through MIDI—the standard

protocol for electronic musical instruments. It consists of a droplet generation system that controls the timing and diameter of droplets using electrically actuated solenoid valves, and a pitch control system that manages resonance at arbitrary frequencies using tubular resonators. This system handles two types of sounds: bubble sounds produced when droplets hit the water surface, and impact sounds produced when droplets hit an object. It realizes a novel musical experience characterized by visual performativity, reproducibility of sound, and physical intervention (bodily interaction) that cannot be achieved through recording playback. While performances of this system have been previously presented, this study provides a detailed explanation of newly developed technologies.

2 Water Droplet Sound Control Mechanism

To enable precise musical expression through water droplet sounds, the system performs physical sound control via the following three sub-mechanisms.

2.1 Droplet Generation Mechanism

Research by Tomita et al. [Tomita et al. 2002], G.J. Frantz [Frantz 1959], and H.C. Pumphrey et al. [Pumphrey and Elmore 1990] has clarified that the bubble sound generated when a water droplet hits a surface is proportional in volume and pitch to the diameter of the bubble created upon impact. The diameter of the resulting bubble depends on the diameter of the water droplet and its velocity at the moment of impact.

By using high-response solenoid valves and controlling the voltage application time (valve open time) in milliseconds, we accurately regulate the droplet diameter and generation timing. Additionally, by fixing the distance between the valve and the water surface at 700 mm, we approximate a constant impact velocity, thereby achieving consistent expression of pitch, volume, and rhythm.



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2.2 Water Pressure Maintenance Mechanism

Variations in water pressure affect the mass and falling velocity of the droplets, which can impair the reproducibility of the pitch. To address this, the system employs a two-stage tank structure. The "Lower Tank", where the valves are installed, is equipped with a water-level sensing mechanism using electrodes. Water is automatically supplied from the "Upper Tank" to maintain a constant water level (and thus constant pressure) in the "Lower Tank". This ensures the stabilization of acoustic characteristics even during long-term performances.

2.3 Pitch Resonance Mechanism

While controlling the droplet diameter allowed for coarse pitch adjustment, strict scalar control was not achieved due to the instability of the water surface caused by continuous droplet impacts. To tune the droplet sounds musically, we introduced a tubular resonator based on the principle of air-column resonance, referencing Kishizuka's [Kishizuka 1991] research on the resonance mechanisms of Suikinkutsu. Kishizuka identified that the resonance in Suikinkutsu follows the principle of air-column resonance, and the natural frequency f within a cylindrical space can be approximated by Equation (1). In this equation, C represents the speed of sound, a is the radius of the cylinder, h is the length of the cylinder, and u_{mk} refers to the coefficient related to vibration modes identified by Kido [Kido 1982]. Since C and u_{mk} are approximated based on the environment, it is possible to amplify specific frequency peaks by varying a and h .

$$f = \frac{C}{2\pi} \sqrt{\left(\frac{u_{mk}}{a}\right)^2 + \left(\frac{n\pi}{h}\right)^2} \quad (1)$$

In this study, a cylindrical resonator ($a = 25$ mm) was placed vertically on the water surface, and droplets were dropped into the water inside the resonator to amplify specific frequency peaks. Figure 2 compares the spectrograms of droplet sounds with and without the resonator. In both cases, the solenoid valve open time was fixed at 80 ms to ensure identical droplet volumes. Figure 2 shows that the use of the resonator results in a distinct peak at approximately 512 Hz with greater sustain. To enable musical performance, we mapped specific values of h to the frequencies of the chromatic scale. By using an actuator (motor fader) to raise and lower the resonator, we control h in real-time, achieving pitch control over a range from E4 to C6.

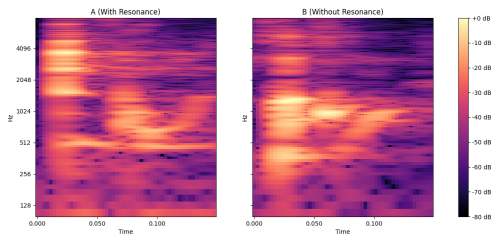


Figure 2: Comparison of spectrograms of droplet sounds with and without the resonator.

3 Performance System

The system processes signals from general-purpose music production software and MIDI devices in real-time. Received MIDI short messages are converted into physical actions through the following process:

- **Channel:** Used to select the module to be played; a parameter for controlling multiple units or other MIDI devices.
- **Note Message (Pitch):** Designates the target height h for the resonator lifting mechanism and triggers droplet generation.
- **Velocity Message (Key Strike Speed):** Assigned to control the valve open time (across 128 levels), adjusting the dynamics and volume by changing the droplet size.
- **Processing:** An Arduino UNO receives these signals and executes the resonator movement and the generation of the droplet sound with a latency of less than 0.4 seconds.

4 Participant Experience

The proposed system provides an instrumental performance experience using acoustic water-droplet sounds. Participants(Performers) can control the droplets in real-time through a MIDI keyboard or music sequencer to play melodies and rhythms. Furthermore, the system allows for physical interventions impossible with digital sound sources, such as muting the sound by intercepting the droplet's path with a hand or altering the timbre by changing the impact objects. This provides a "water performance" where visual and auditory elements are perfectly synchronized.

Acknowledgments

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